

REMEMBER FORWARD THE PATIENT MESSAGE

Container Specifications, Plate Tiering & Audio Preservation Architecture

Version 2.0 — Updated with Schedule 80 Standard & Revised Language Preservation Architecture

VERSION 2.0 CHANGES: (1) Schedule 80 PVC upgraded to standard across all tier designs. (2) Plate tiering architecture revised — content depth now independent of container type. (3) Language preservation series expanded from 1 plate to 4 plates per language plus audio disc. (4) Audio disc specifications added. (5) Playback component preservation guidance added.

Section 1 — Container Specifications (Updated v2.0)

CRITICAL UPDATE: Schedule 80 PVC is now the standard recommendation for all capsule builds. The additional cost is \$15–\$30 per capsule. Over a 2,000-year burial timeline this upgrade is unambiguously worth it. Do not use Schedule 40 if Schedule 80 is available in your area.

1.1 Why Schedule 80 Instead of Schedule 40

Both Schedule 40 and Schedule 80 PVC use the same chemical compound and have the same outer diameter. The difference is wall thickness. Schedule 80 walls are approximately 60% thicker, providing significantly higher crush resistance, impact resistance, and long-term stress tolerance under buried conditions.

Property	Schedule 40 (OLD standard)	Schedule 80 (NEW standard)
Wall thickness (4" pipe)	~6mm	~9.5mm
Pressure rating (4" pipe)	220 psi	320 psi
Crush resistance	Moderate	High
Cost (2ft section, 4")	\$8–\$15	\$15–\$25
Additional capsule cost	Baseline	~\$15–\$30 more
2,000-year safety margin	Adequate	Recommended

The soil pressure on a buried capsule at 4 to 6 feet depth is approximately 60 to 90 pounds per square foot. Over centuries, freeze-thaw cycles, root intrusion, soil compaction, and seismic activity add cumulative stress. Schedule 80 provides a substantially larger safety margin for the long-term survival of the capsule under these conditions.

1.2 Thread Sealant Upgrade (All Tiers)

UPDATED RECOMMENDATION: Use Teflon tape PLUS marine-grade epoxy on all threaded joints, not standard pipe dope or Teflon tape alone. Marine epoxy is chemically inert, sets permanently, and resists the full range of soil chemistry including acidic peat soils and alkaline limestone soils that degrade standard sealants over decades.

Application method: wrap threads with Teflon tape first (3 to 4 wraps), then apply a thin bead of marine epoxy over the Teflon before threading the cap. Allow 24 hours cure time before burial. The combination

creates a seal that is both mechanically strong (Teflon) and chemically resistant (epoxy).

1.3 Tier A — PVC Buried Capsule (Revised)

Household build · Schedule 80 PVC · \$75–\$150 · Target lifespan 500–1,000 years

Materials List (Updated)

MATERIAL: 4-inch Schedule 80 PVC pipe, 18–24 inch section

MATERIAL: 4-inch Schedule 80 threaded end caps (2x)

MATERIAL: 4-inch Schedule 80 threaded adapter (if pipe not pre-threaded)

SEALANT: Teflon tape (plumber's tape)

SEALANT: Marine-grade epoxy (e.g. West System, Loctite Marine)

DESICCANT: Food-grade silica gel desiccant packets (6–10 packets)

DESICCANT: Oxygen absorbers (300–500cc capacity, 4–6 packets)

PACKAGING: Mylar vacuum-seal bags for plate stacks

PACKAGING: Acid-free tissue paper between plates

Assembly Sequence

- Clean all pipe surfaces and threads with isopropyl alcohol. Allow to dry completely.
- Seal one end cap using Teflon tape + marine epoxy. Allow 24 hours to cure.
- Wrap plate stacks in acid-free tissue, seal in Mylar bags with oxygen absorbers.
- Place desiccant packets at bottom of pipe.
- Insert plate stacks vertically or horizontally depending on plate orientation.
- Place remaining desiccant packets above plate stack.
- Seal second end cap with Teflon tape + marine epoxy.
- Allow 24 hours full cure before burial.
- Burial depth: minimum 4 feet, ideally 5–6 feet, below local frost line.
- Orientation: horizontal preferred to distribute soil pressure across length rather than ends.

Optional Upgrades

- Wrap entire sealed pipe in 2–3 layers of self-amalgamating silicone tape for additional moisture barrier
- Place pipe inside a larger Schedule 80 pipe section as a secondary containment layer
- Surround with clean gravel before backfilling to improve drainage and reduce direct soil contact
- Mark burial location with a non-ferrous metal stake driven 12 inches above capsule

1.4 Tier B — Titanium Sphere (Unchanged except sealant upgrade)

Premium build · Grade 5 Ti-6Al-4V · \$350–\$600 · Target lifespan 10,000+ years

The argon atmosphere fill and silica aerogel insulation specifications are unchanged. The sealing specification is NOT: Viton is better than Buna-N but no elastomer has a validated multi-century life, so this tier's stated 10,000-year figure applies to the shell and not to the seal. Ranked by demonstrated life: welded or diffusion-bonded metal (indefinite) > cast lead (2,000+ yrs, Roman) = pitch on ceramic (2,000-4,000 yrs) > glass-to-metal compression > metal C-ring > elastomer O-ring. The two lowest-technology options beat every modern polymer.

UNCHANGED: Inner chamber: octahedron frame, argon atmosphere, silica aerogel insulation

UNCHANGED: Pyramid cavity at center, apex at 51.83° Giza angle

UPDATED: O-ring: Viton (FKM) outlasts Buna-N, but rate this closure at 100-300 years, not 10,000. Weld it if it must last longer.

CONFIRMED: CORRECTED 2026-08-20: a 6-inch sphere holds 1.85 litres. A 206-plate titanium stack is roughly 3.4 litres and 15 kg, so it does NOT fit. Real capacity is about 100-120 thin plates. Size the vessel to the plate set, or the plate set to the vessel — but measure first.

1.5 Tier C — Ocean Survival Capsule (Unchanged except sealant)

Ocean deployment · HDPE hull + Ti pressure vessel · \$400–\$900 · Rated to 600 atm

Ocean deployment now uses titanium plates, which are inert in seawater and need no plating; the gold-plated nickel requirement belonged to the nickel design and is withdrawn. Vitrified stoneware is also viable in seawater given impact protection. Audio discs should use DMM copper with gold plating.

1.6 Superior Option: Borosilicate Glass + Titanium Shell

Previously identified as superior for ocean and acid environments. This combination remains the highest-specification option for extreme long-term survival. The borosilicate glass inner sphere provides chemical inertness that no metal matches, and the titanium outer shell provides mechanical protection. Recommended for institutional or high-value deployments where cost is secondary to survival probability.

Section 2 — Revised Plate Tiering Architecture (v2.0)

CRITICAL REVISION: The previous architecture linked plate depth to container type. This was incorrect. Plate set depth is now independent of container type. Any container can hold any plate set. The tiering is purely about content depth and cost of the plates themselves, not the container they go into.

2.1 The Four Language Plate Series (Per Language)

Each language now has a four-plate preservation series. These four plates together constitute genuine language preservation rather than a minimal decipherment aid. The plates are designed to be used together but each adds independent value.

Plate	Name	Contents	What It Preserves
Plate A	The Script Plate	Complete alphabet or syllabary · Gematria or numerical values · Script direction · Letter names · Mayan dot-bar number bridge · Core vocabulary 12–16 words · Bridge phrase	Writing system · Basic decipherment key · Numerical system

Plate	Name	Contents	What It Preserves
Plate B	The Phonology Plate	Every phoneme with articulation diagrams showing mouth position and tongue placement · Tonal contour diagrams for tonal languages · Click consonant diagrams for click languages · Vowel chart · Minimal pairs demonstrating sound contrasts	Sound system · Pronunciation · The critical dimension plates alone cannot convey
Plate C	The Grammar & Vocabulary Plate	Swadesh 207-word core vocabulary list · Basic sentence structures · Question formation · Negation · Tense/aspect marking · Number system · Key grammatical patterns	Grammar · Vocabulary corpus sufficient for basic reconstruction
Plate D	The Running Text Plate	Bridge phrase expanded to full paragraph · Traditional story or prayer · UN Declaration of Human Rights Preamble (translated into 500+ languages — serves as universal parallel text for comparative linguistics)	Natural language in use · Parallel text anchor connecting to every other preservation project

2.2 The Three Plate Set Tiers

Any plate set can go in any container. The tiers below describe content depth and cost, not container type.

Tier 1 — Minimal Preservation Set

Plate A only per language · 56 plates total · ~\$840–\$1,680 for full set

Contents: 6 knowledge plates + Plate A for all 50 languages. Provides a decipherment aid for each language and the complete pictographic knowledge archive. A brilliant finder with access to other surviving materials can use this to decode languages. Best for maximum distribution at minimum cost.

Component	Count	Est. Cost per Plate	Est. Total
Knowledge plates (1–6)	6	\$15–\$30	\$90–\$180
Language Plate A × 50 languages	50	\$15–\$30	\$750–\$1,500
Audio disc (optional)	1–6	\$50–\$100	\$50–\$600
TOTAL (without audio)	56	—	\$840–\$1,680
TOTAL (with audio)	56 + discs	—	\$890–\$2,280

Tier 2 — Standard Preservation Set

Plates A + B per language · 106 plates total · ~\$1,590–\$3,180 for full set

Contents: 6 knowledge plates + Plates A and B for all 50 languages. Adds the phonology plate — the single most critical missing piece in Tier 1. A finder can now reconstruct both the writing system and the sound system of each language. Recommended as the standard for most builders who can afford it.

Component	Count	Est. Cost per Plate	Est. Total
Knowledge plates (1–6)	6	\$15–\$30	\$90–\$180
Language Plates A+B × 50 languages	100	\$15–\$30	\$1,500–\$3,000
Audio disc (strongly recommended at this tier)	1–6	\$50–\$100	\$50–\$600
TOTAL (without audio)	106	—	\$1,590–\$3,180
TOTAL (with audio)	106 + discs	—	\$1,640–\$3,780

Tier 3 — Complete Preservation Set

All four plates per language + audio discs · 206 plates + discs · ~\$3,090–\$6,780

Contents: 6 knowledge plates + all four language plates for all 50 languages + audio discs. This is genuine full-spectrum language preservation. A finder can reconstruct the writing system, sound system, vocabulary, grammar, and natural usage of each language entirely from the capsule contents alone, with no other surviving materials required. The gold standard for the project.

Component	Count	Est. Cost per Plate	Est. Total
Knowledge plates (1–6 plus new plates)	8+	\$15–\$30	\$120–\$240
Language Plates A+B+C+D × 50 languages	200	\$15–\$30	\$3,000–\$6,000
Audio discs (all 50 languages, 4–6 discs)	4–6	\$50–\$500	\$200–\$3,000
Audio playback guide plate	1	\$15–\$30	\$15–\$30
TOTAL (titanium-engraved audio)	206+	—	\$3,090–\$6,330
TOTAL (DMM gold-plated audio — ocean spec)	206+	—	\$5,000–\$10,000+

2.3 Physical Dimensions of Each Tier

Stack heights below assume the old 1.5 mm nickel plate. Titanium plates run 0.5-0.8 mm, so a metal stack is roughly half the height shown - but check the vessel volume as well as the height, because the 6-inch sphere claim above did not survive that check. Ceramic relief tiles are 8 mm minimum and do not belong in these figures at all.

Tier	Plate Count	Stack Height	Fits in
Tier 1 (Plate A only)	56 plates	84mm (3.3 inches)	Any PVC or sphere capsule
Tier 2 (Plates A+B)	106 plates	159mm (6.3 inches)	4-inch PVC 12"+ or any sphere
Tier 3 (All four plates)	206 plates	309mm (12.2 inches)	4-inch PVC 18"+ or any sphere
Tier 3 + audio discs	206 + 6 discs	~345mm (13.6 inches)	4-inch PVC 20"+ or any sphere

2.4 Recommended Combinations

These are suggested starting points. Any combination is valid. The most important thing is that capsules get built and buried, not that every capsule reaches Tier 3 specification.

Builder Profile	Recommended Combination	Rationale
Household builder, modest budget	Schedule 80 PVC + Tier 1 plate set	Maximum distribution at minimum cost. Get the knowledge out into the world.
Household builder, can save for it	Schedule 80 PVC + Tier 2 plate set + audio disc	Adds phonology and audio — the two most critical missing pieces in Tier 1.
Serious prepper or community group	Titanium sphere + Tier 3 plate set + audio discs	Full preservation corpus. The gold standard household build.
Institutional or educational	Borosilicate/Ti shell + Tier 3 + DMM gold audio	Maximum survival specification. For permanent archival installation.
Ocean deployment	Ocean capsule + Tier 3 + DMM gold-plated audio	Gold-plated plates mandatory. DMM audio for maximum longevity.
Mass distribution project	Schedule 80 PVC + Tier 1 × many units	Distribute hundreds of Tier 1 capsules rather than a few Tier 3. Redundancy wins.

Section 3 — Audio Disc Specifications

The audio disc is a companion artifact to the language plate series. It preserves the sound dimension of each language that no engraved plate can fully convey — tones, clicks, retroflex consonants, natural speech rhythm, and the human voice itself.

3.1 The Three Audio Disc Options

Option	Method	Fidelity	Survival	Cost per Disc	Best For
Option A: Laser-engraved titanium	Fiber laser cuts audio grooves directly into titanium plate at the same service as the knowledge plates	Lo-fi: 4–5 bit, ~4.5kHz. Speech intelligible, music degraded	Same as titanium plates — centuries to millennia under ideal conditions	\$50–\$100	Tier 1 and 2 capsules. Integrated manufacturing with plates.
Option B: DMM copper + gold plate	Direct Metal Mastering lathe cuts into copper. Electroplated with nickel then gold.	Hi-fi: full frequency response, equivalent to professional vinyl	Indefinite. Gold does not corrode. Copper alloy extremely stable.	\$300–\$500	Tier 3 and ocean capsules. Maximum preservation specification.
Option C: Lathe-cut polycarbonate	Embossing stylus cuts grooves into polycarbonate disc	Good: 20Hz–14kHz, low surface noise	Decades only. Not suitable for long-term burial.	\$12–50	Prototyping and testing audio content only. Not for final capsules.

3.2 Audio Content Specification

One audio disc set serves the entire language series. At 16■ RPM (the Voyager standard, chosen to maximize duration), a 12-inch disc holds approximately 16–18 minutes per side. Six double-sided discs hold approximately 192–216 minutes — sufficient for 50 languages at approximately 4 minutes per language.

Per-Language Audio Content (4 minutes per language)

- Phoneme demonstration set — every phoneme in isolation and in minimal pairs (60–90 seconds)
- Number sequence spoken aloud 1–20, bridging to Mayan dot-bar visual system (20–30 seconds)
- Core vocabulary spoken — the 16 words from Plate A spoken clearly with a pause between each (40–60 seconds)
- Running text reading — the bridge phrase plus the UN Declaration preamble from Plate D read at natural pace (60–90 seconds)
- Tonal demonstration if applicable — all tones of the language demonstrated with minimal pairs (20–30 seconds, tonal languages only)

Universal Audio Content (on first disc, before language series)

- The harmonic series — pure tones at mathematically related frequencies demonstrating acoustic physics (60 seconds)
- A universal tone reference — 440 Hz A4, the international tuning standard, held for 10 seconds
- Playback speed calibration signal — a known frequency tone allowing finder to calibrate rotation speed
- The bridge phrase in English as anchor reference (30 seconds)

3.3 Playback Component Preservation

Include these non-electronic playback components sealed separately within the capsule. They require no moving parts and will survive 2,000 years without question.

INCLUDE: Pre-formed gold or platinum stylus tip (needle) — a small pointed rod 0.5mm diameter, sealed in a glass vial

INCLUDE: Pre-formed gold sheet cone amplifier — folded into horn shape, approximately 6 inches long, sealed flat

INCLUDE: Audio Playback Guide plate (Plate 32) — showing needle construction, tone arm assembly, rotation speed calibration using pendulum, and cone horn assembly

The playback guide plate is the most critical component for audio utility. Without it, a finder cannot know the disc contains audio, cannot determine the correct playback method, and cannot calibrate the rotation speed. The guide plate must be included whenever an audio disc is included.

3.4 Disc Numbering and Labeling

Each disc face must be engraved (not printed) with the following information in the margin area outside the groove spiral:

- Disc number and total (e.g. DISC 1 OF 6)
- Languages contained on this disc face (e.g. ENGLISH · SPANISH · PORTUGUESE · FRENCH)

- Rotation direction arrow
- Starting groove indicator (arrow pointing to outermost groove)
- Playback speed in revolutions per minute (16■ RPM)
- Reference to the Audio Playback Guide plate number

Section 4 — Updated Complete Plate Index

This is the master plate index reflecting the revised architecture. Total plates in complete Tier 3 set: 214 plates plus 6 audio discs.

4.1 Knowledge Plates (1–8)

Plate	Title	Status	Notes
1	Rosetta Index — pictographic key, Mayan numbers, star map	COMPLETE v3	Verified clean, no overlaps
2	Water · Fire · Shelter	COMPLETE v2	Portrait, Plate 1 style
3	Agriculture & Seeds	COMPLETE v2	Portrait, Plate 1 style
4	Technology Re-foundation	COMPLETE v2	Iron, glass, math, printing
5	Governance & Ethics	COMPLETE v2	Corruption sequence, golden rule, rights
6	The Ten Concords & Star Navigation	COMPLETE v2	Ten concords + star charts
7	Encoding & C ommunication Principles	PENDING	Morse, binary, dot-bar, encoding stack diagram
8	Audio Playback Guide	PENDING	Needle construction, tone arm, rotation calibration

4.2 Language Plates — Hebrew Series (Plates 9–16)

Plates	Language	Status	Notes
9A–9D	Biblical Hebrew (Pale o-Hebrew / K'tav Ivri)	PENDING	Pre-exile script, Torah tradition

Plates	Language	Status	Notes
10A–10D	Mishnaic Hebrew (Square script + nikkud)	PENDING	K'tav Ashuri, vowel points, Talmudic
11A–11D	Yemenite Hebrew	PENDING	Distinct script tradition, ancient pronunciation
12A–12D	Modern Hebrew (Ivrit)	PLATE A COMPLETE	Prototype. Plates B, C, D pending.

4.3 Language Plates — Full Series (50 Languages)

Each language has four plates: A (Script), B (Phonology), C (Grammar & Vocabulary), D (Running Text). Status shown for Plate A. Plates B, C, D pending for all languages.

Plates	Language	Region	Script	Priority
13A–13D	Ancient Egyptian (Hieroglyphic)	Middle East / Africa	Hieroglyphic	High — already decoded once
14A–14D	Sumerian / Akkadian (Cuneiform)	Middle East	Cuneiform	High — oldest writing system
15A–15D	Tamil	South Asia	Tamil script	High — oldest continuous literature
16A–16D	Sanskrit	South Asia	Devanagari	High — root of Indo-European family
17A–17D	Bengali	South Asia	Bengali script	High — 230M speakers
18A–18D	Mandarin Chinese	East Asia	Logographic + Pinyin	High — 1.1B speakers
19A–19D	Japanese	East Asia	Hiragana + Katakana + Kanji	High — 3-script system
20A–20D	Malay / Indonesian	SE Asia	Latin	High — 260M speakers, archipelago
21A–21D	Arabic (Classical & Modern)	Middle East / Africa	Arabic script	High — 400M speakers, right-to-left
22A–22D	Russian (Cyrillic)	Eurasia	Cyrillic	High — largest geographic coverage
23A–23D	English	Global	Latin	High — current global lingua franca
24A–24D	Spanish	Americas / Global	Latin	High — 500M speakers
25A–25D	Portuguese	Americas / Africa	Latin	High — 260M speakers, Brazil
26A–26D	Georgian (Mkhedruli)	Caucasus	Mkhedruli	High — unique independent script
27A–27D	Swahili	East Africa	Latin / Arabic	High — 200M speakers, lingua franca
28A–28D	Amharic (Ge'ez script)	Horn of Africa	Ge'ez syllabary	High — ancient Christian tradition

Plates	Language	Region	Script	Priority
29A–29D	Hausa	West Africa	Latin / Ajami	High — 100M speakers
30A–30D	Yoruba	West Africa	Latin	High — 50M speakers, tonal
31A–31D	Zulu / Nguni	Southern Africa	Latin	High — click consonants, oldest language branch
32A–32D	Quechua	Andes / Americas	Latin	High — Inca civilization
33A–33D	Nahuatl	Mesoamerica	Latin	High — Aztec civilization
34A–34D	Farsi / Persian	Central Asia	Arabic-derived	Medium — 110M speakers, great literary tradition
35A–35D	Turkish	Eurasia	Latin (since 1928)	Medium — 88M speakers
36A–36D	Vietnamese	SE Asia mainland	Latin + tones	Medium — 96M speakers, tonal
37A–37D	Punjabi	South Asia	Gurmukhi	Medium — 125M speakers, Sikhism
38A–38D	Tagalog / Filipino	Pacific	Latin	Medium — covers Oceania
39A–39D	Tibetan	Himalaya	Tibetan script	Medium — unique script, contemplative tradition
40A–40D	Mongolian	Central Asia	Traditional Mongolian	Medium — vertical script, steppe civilization
41A–41D	Igbo	West Africa	Latin	Medium — 45M speakers, Nigeria
42A–42D	Somali	Horn of Africa	Latin	Medium — 25M speakers
43A–43D	Maori / Polynesian	Pacific	Latin	Medium — only Pacific language
44A–44D	Greek (Ancient / Modern)	Europe	Greek alphabet	Medium — foundation of Western science
45A–45D	Hindi	South Asia	Devanagari	Medium — 600M speakers
46A–46D	Swahili (Ajami)	East Africa	Arabic-derived	Lower — historical bridge script
47–50	Reserved for community submissions	Global	Various	TBD — open-source contributions

Remember Forward — The Patient Message

Buy it. Build it. Bury it. For someone you will never meet.

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Corrected 2026-09-01. The errata of 2026-08-20 revised this document's prose but left its specification tables unchanged, so the two contradicted each other. See `containers/ERRATA.md` and `AUDIT_2026-09-01.md`. The audio section had been missed entirely by the substrate reversal.